로컬 repo 구성하기

1. 디스크 연결

100G 연결 > LVM 형태로 구현

```
vm에서 100G 디스크 연결, 이후 부팅
```

[root@workstation ~]# lsblk

```
NAME   MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda      8:0    0  100G  0 disk
 ├──sda1   8:1    0    2G  0 part /boot
 ├──sda2   8:2    0   20G  0 part /var/log
 ├──sda3   8:3    0   10G  0 part /var
 ├──sda4   8:4    0    1K  0 part
 ├──sda5   8:5    0    8G  0 part [SWAP]
 ├──sda6   8:6    0    5G  0 part /home
 └──sda7   8:7    0   55G  0 part /
sdb      8:16   0  100G  0 disk
sr0     11:0    1 1024M  0 rom
```

[root@workstation ~]# fdisk /dev/sdb

```
Welcome to fdisk (util-linux 2.32.1).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS disklabel with disk identifier 0xb6a9555a.

Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p):

Using default response p.
Partition number (1-4, default 1):
First sector (2048-209715199, default 2048):
Last sector, +sectors or +size{K,M,G,T,P} (2048-209715199, default 209715199):

Created a new partition 1 of type 'Linux' and of size 100 GiB.

Command (m for help): p
Disk /dev/sdb: 100 GiB, 107374182400 bytes, 209715200 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xb6a9555a

Device     Boot Start       End   Sectors  Size Id Type
/dev/sdb1        2048 209715199 209713152  100G 8e Linux LVM

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

d
```

[root@workstation ~]# lsblk

```
NAME   MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda      8:0    0  100G  0 disk
 ├──sda1   8:1    0    2G  0 part /boot
 ├──sda2   8:2    0   20G  0 part /var/log
 ├──sda3   8:3    0   10G  0 part /var
 ├──sda4   8:4    0    1K  0 part
 ├──sda5   8:5    0    8G  0 part [SWAP]
 ├──sda6   8:6    0    5G  0 part /home
 └──sda7   8:7    0   55G  0 part /
sdb      8:16   0  100G  0 disk
 └──sdb1   8:17   0  100G  0 part
sr0     11:0    1 1024M  0 rom
```

현재 LVM 이 설치되어 있지 않다.

[root@workstation ~]# rpm -qa | grep lvm

```
```

[root@workstation ~]# yum -y install lvm2

```
Last metadata expiration check: 18:26:48 ago on Wed 30 Mar 2022 03:19:16 AM EDT.
Dependencies resolved.
================================================
 Package
       Arch   Version              Repo    Size
================================================

Installed:
  device-mapper-event-8:1.02.177-11.el8_5.x86_64
  device-mapper-event-libs-8:1.02.177-11.el8_5.x86_64
  device-mapper-persistent-data-0.9.0-4.el8.x86_64
  libaio-0.3.112-1.el8.x86_64
  lvm2-8:2.03.12-11.el8_5.x86_64
  lvm2-libs-8:2.03.12-11.el8_5.x86_64

Complete!
```

[root@workstation ~]# rpm -ql lvm2

[root@workstation ~]# pvcreate /dev/sdb1

```
Physical volume "/dev/sdb1" successfully created.
```

[root@workstation ~]# vgcreate data /dev/sdb1

```
Volume group "data" successfully created
```

[root@workstation ~]# lvcreate -n repo -l 100%FREE data

```
Logical volume "repo" created.
```

[root@workstation ~]# mkfs.xfs /dev/data/repo

```
meta-data=/dev/data/repo         isize=512    agcount=4, agsize=6553344 blks
         =                       sectsz=512   attr=2, projid32bit=1
         =                       crc=1        finobt=1, sparse=1, rmapbt=0
         =                       reflink=1
data     =                       bsize=4096   blocks=26213376, imaxpct=25
         =                       sunit=0      swidth=0 blks
naming   =version 2              bsize=4096   ascii-ci=0, ftype=1
log      =internal log           bsize=4096   blocks=12799, version=2
         =                       sectsz=512   sunit=0 blks, lazy-count=1
realtime =none                   extsz=4096   blocks=0, rtextents=0
```

[root@workstation ~]# mkdir /REPO

[root@workstation ~]# mkdir /REPO

[root@workstation ~]# mount /dev/data/repo /REPO

[root@workstation ~]# df

```
Filesystem            1K-blocks    Used Available Use% Mounted on
devtmpfs                 383484       0    383484   0% /dev
tmpfs                    400500       0    400500   0% /dev/shm
tmpfs                    400500   10832    389668   3% /run
tmpfs                    400500       0    400500   0% /sys/fs/cgroup
/dev/sda7              57639428 1837312  55802116   4% /
```

```
/dev/sda3               10475520  397260  10078260   4% /var
/dev/sda6                5232640   69556   5163084   2% /home
/dev/sda1                2086912  280688   1806224  14% /boot
/dev/sda2               20961280  192712  20768568   1% /var/log
tmpfs                      80100       0     80100   0% /run/user/0
/dev/mapper/data-repo 104802308  763724 104038584   1% /REPO
```

[root@workstation ~]# umount /REPO

[root@workstation ~]# vi /etc/fstab

```
UUID=8a2af94c-a3a1-4d33-bc62-f98b416cb7fb /REPO                   xfs     defaults
  0 0
```

[root@workstation ~]# mount -a

[root@workstation ~]# df

[root@workstation ~]# yum -y install yum-utils createrepo

```
Last metadata expiration check: 18:40:24 ago on Wed 30 Mar 2022 03:19:16 AM EDT.
Dependencies resolved.
================================================================================
 Package                 Architecture  Version                  Repository        Size
================================================================================
Installing:
 createrepo_c            x86_64        0.17.2-3.el8             appstream         87 k
 yum-utils               noarch        4.0.21-4.el8_5           baseos            71 k
Installing dependencies:
 createrepo_c-libs       x86_64        0.17.2-3.el8             appstream        114 k
 drpm                    x86_64        0.4.1-3.el8              appstream         67 k

Transaction Summary
================================================================================
Install  4 Packages

Total download size: 340 k
Installed size: 598 k
Downloading Packages:
(1/4): createrepo_c-0.17.2-3.el8.x86_64.rpm            158 kB/s |  87 kB     00:00
(2/4): createrepo_c-libs-0.17.2-3.el8.x86_64.rpm       204 kB/s | 114 kB     00:00
(3/4): drpm-0.4.1-3.el8.x86_64.rpm                     116 kB/s |  67 kB     00:00
(4/4): yum-utils-4.0.21-4.el8_5.noarch.rpm             207 kB/s |  71 kB     00:00
--------------------------------------------------------------------------------
Total                                                  120 kB/s | 340 kB     00:02
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                        1/1
  Installing       : drpm-0.4.1-3.el8.x86_64                                1/4
  Installing       : createrepo_c-libs-0.17.2-3.el8.x86_64                  2/4
  Installing       : createrepo_c-0.17.2-3.el8.x86_64                       3/4
  Installing       : yum-utils-4.0.21-4.el8_5.noarch                        4/4
  Running scriptlet: yum-utils-4.0.21-4.el8_5.noarch                        4/4
  Verifying        : createrepo_c-0.17.2-3.el8.x86_64                       1/4
  Verifying        : createrepo_c-libs-0.17.2-3.el8.x86_64                  2/4
  Verifying        : drpm-0.4.1-3.el8.x86_64                                3/4
  Verifying        : yum-utils-4.0.21-4.el8_5.noarch                        4/4
```

```
Installed:
  createrepo_c-0.17.2-3.el8.x86_64          createrepo_c-libs-0.17.2-3.el8.x86_64
  drpm-0.4.1-3.el8.x86_64                   yum-utils-4.0.21-4.el8_5.noarch

Complete!
```

[root@workstation ~]# cd /REPO/

```
[root@workstation REPO]#
```

[root@workstation Packages]# cd

[root@workstation ~]# mkdir bin

[root@workstation ~]# cd bin

[root@workstation bin]# yum repolist | grep -v 'repo name' | awk '{print $1}' > list

[root@workstation bin]# cat list

```
appstream
baseos
epel
epel-modular
extras
powertools
```

[root@workstation bin]# vi repo_import.sh

```
#!/usr/bin/bash
for list in $(cat /root/bin/list)
do
reposync -p /REPO -m -n --download-metadata --repoid=$list
done
```

[root@workstation bin]# chmod 500 repo_import.sh

[root@workstation REPO]# repo_import.sh

[root@workstation repo]# yum -y install nginx

```
Last metadata expiration check: 0:06:13 ago on Thu Mar 31 16:38:13 2022.
Dependencies resolved.
================================================================================
 Package                      Arch   Version                Repo       Size
================================================================================
Installing:
 nginx                        x86_64 1:1.14.1-9.module+el8.4.0+542+81547229
                                                            appstream 566 k
Installed:
  dejavu-fonts-common-2.35-7.el8.noarch
  dejavu-sans-fonts-2.35-7.el8.noarch
  fontconfig-2.13.1-4.el8.x86_64
  fontpackages-filesystem-1.44-22.el8.noarch
  gd-2.2.5-7.el8.x86_64
  jbigkit-libs-2.1-14.el8.x86_64
  libX11-1.6.8-5.el8.x86_64
  libX11-common-1.6.8-5.el8.noarch
  libXau-1.0.9-3.el8.x86_64
  libXpm-3.5.12-8.el8.x86_64
  libjpeg-turbo-1.5.3-12.el8.x86_64
  libtiff-4.0.9-20.el8.x86_64
  libwebp-1.0.0-5.el8.x86_64
  libxcb-1.13.1-1.el8.x86_64
  nginx-1:1.14.1-9.module+el8.4.0+542+81547229.x86_64
  nginx-all-modules-1:1.14.1-9.module+el8.4.0+542+81547229.noarch
  nginx-filesystem-1:1.14.1-9.module+el8.4.0+542+81547229.noarch
  nginx-mod-http-image-filter-1:1.14.1-9.module+el8.4.0+542+81547229.x86_64
  nginx-mod-http-perl-1:1.14.1-9.module+el8.4.0+542+81547229.x86_64
  nginx-mod-http-xslt-filter-1:1.14.1-9.module+el8.4.0+542+81547229.x86_64
  nginx-mod-mail-1:1.14.1-9.module+el8.4.0+542+81547229.x86_64
  nginx-mod-stream-1:1.14.1-9.module+el8.4.0+542+81547229.x86_64
  perl-Carp-1.42-396.el8.noarch
  perl-Data-Dumper-2.167-399.el8.x86_64
  perl-Digest-1.17-395.el8.noarch
  perl-Digest-MD5-2.55-396.el8.x86_64
  perl-Encode-4:2.97-3.el8.x86_64
```

```
  perl-Errno-1.28-420.el8.x86_64
  perl-Exporter-5.72-396.el8.noarch
  perl-File-Path-2.15-2.el8.noarch
  perl-File-Temp-0.230.600-1.el8.noarch
  perl-Getopt-Long-1:2.50-4.el8.noarch
  perl-HTTP-Tiny-0.074-1.el8.noarch
  perl-IO-1.38-420.el8.x86_64
  perl-IO-Socket-IP-0.39-5.el8.noarch
  perl-IO-Socket-SSL-2.066-4.module+el8.4.0+512+d4f0fc54.noarch
  perl-MIME-Base64-3.15-396.el8.x86_64
  perl-Mozilla-CA-20160104-7.module+el8.4.0+529+e3b3e624.noarch
  perl-Net-SSLeay-1.88-1.module+el8.4.0+512+d4f0fc54.x86_64
  perl-PathTools-3.74-1.el8.x86_64
  perl-Pod-Escapes-1:1.07-395.el8.noarch
  perl-Pod-Perldoc-3.28-396.el8.noarch
  perl-Pod-Simple-1:3.35-395.el8.noarch
  perl-Pod-Usage-4:1.69-395.el8.noarch
  perl-Scalar-List-Utils-3:1.49-2.el8.x86_64
  perl-Socket-4:2.027-3.el8.x86_64
  perl-Storable-1:3.11-3.el8.x86_64
  perl-Term-ANSIColor-4.06-396.el8.noarch
  perl-Term-Cap-1.17-395.el8.noarch
  perl-Text-ParseWords-3.30-395.el8.noarch
  perl-Text-Tabs+Wrap-2013.0523-395.el8.noarch
  perl-Time-Local-1:1.280-1.el8.noarch
  perl-URI-1.73-3.el8.noarch
  perl-Unicode-Normalize-1.25-396.el8.x86_64
  perl-constant-1.33-396.el8.noarch
  perl-interpreter-4:5.26.3-420.el8.x86_64
  perl-libnet-3.11-3.el8.noarch
  perl-libs-4:5.26.3-420.el8.x86_64
  perl-macros-4:5.26.3-420.el8.x86_64
  perl-parent-1:0.237-1.el8.noarch
  perl-podlators-4.11-1.el8.noarch
  perl-threads-1:2.21-2.el8.x86_64
  perl-threads-shared-1.58-2.el8.x86_64

Complete!
```

[root@workstation repo]# vi /etc/nginx/nginx.conf

```
    server {
        listen       80 default_server;
        listen       [::]:80 default_server;
        server_name  _;
        root         /REPO ;

        location / {
                autoindex on;
        }
```

[root@workstation repo]# systemctl restart nginx

[root@workstation repo]# firewall-cmd --permanent --add-service=http

success

[root@workstation repo]# firewall-cmd --reload

success

[root@workstation repo]# yum -y install policycoreutils-python-utils

[root@workstation repo]# semanage fcontext -a -t httpd_sys_content_t '/REPO(/.*)?'

[root@workstation repo]# restorecon -Rv /REPO

[root@workstation ~]# cd

[root@workstation ~]# cd bin

[root@workstation bin]# vi repo_DB.sh

```
#!/usr/bin/bash
createrepo /REPO/appstream/
createrepo /REPO/baseos/
createrepo /REPO/epel/
createrepo /REPO/epel-modular/
createrepo /REPO/extras/
```

```
createrepo /REPO/powertools/
```

추가 작업 server1-4 에 등록할 repo 파일 구성

```
[root@workstation repo]# ls
local-appstream.repo
local-baseos.repo
local-epel-modular.repo
local-epel.repo
local-extras.repo
local-powertools.repo


[root@workstation repo]# cat local-appstream.repo
[local-appstream]
name=local-appstream
baseurl=http://workstation.example.com/appstream/
enabled=1
gpgcheck=0


동일한 방식으로 웹페지의 index 로 출력되는 디렉토리를 baseurl 마지막부분에 등록하여 repo 파일을 생성한다.

다음 작업을 각각의 서버에서 진행한다.

cd /etc/yum.repos.d/
rm -rf *
scp root@workstation:/root/repo/*.repo .
yum clean all
yum repolist
yum list
```